Paras Verma, Ph.D. (He/Him)

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RESEARCH PROFILE AND SKILLS

About: Computational biologist focused on transcriptomics, proteogenomics, structural dynamics, and multi-omics ML interpretation.

Most transcriptomic pipelines focus on differential expression or isoform usage. My work is positioned one step downstream: I evaluate whether transcript alterations remodel coding sequence, protein features and peptide evidence, with a broader scope to understand pathogenicity. This makes the analysis useful for selecting biologically functional events for experimental assessment. To do this, I developed ENACT and ENACTdb to classify exon and transcript differences across isoforms and connect them to coding regions, protein features, and proteoform diversity. I contributed also to PEXMap to map mass-spectrometry peptides directly to exons and splice junctions, and an explainable AI pipeline to prioritize ALS biomarkers.

Technical skills

- **Transcriptomics and annotation:** RefSeq, Ensembl, gene and transcript models, exon and isoform classification, *cis*-regulatory splicing consequence interpretation.
- **Proteogenomics, Structural dynamics and protein-aware interpretation:** peptide-to-transcript k-mer mapping PEXMap), exon-exon junction interpretation, domain and IDR/motif mapping, proteoform analysis, atomistic and steered molecular dynamics (MD) simulations, dimensionality reduction approaches, and residue contact/correlation networks.
- **Multi-omics and ML analysis:** cross-tissue meta-transcriptomics, SHAP-decomposed ensemble classifiers, BRCA1/2 variant interpretation, transcriptome-to-protein reasoning, de-novo genome annotation, comparative interpretation across public datasets.
- **Programming, HPC and resources:** Python, R, Bash scripting, annotation workflows, Django 4, MySQL, ReactJS, full-stack database development, Linux HPC cluster handling (Slurm, PBS), and NAS storage management (Synology, Netgear).

WORK EXPERIENCE

Plaksha University – Mohali, India | **Research Associate (Postdoc)** [24/09/2025 - Current]

I am extending the previously developed ENACT framework (detailed in PhD work) to variant interpretation and proteogenomics. I am curating and interpreting population-stratified BRCA variants with attention to annotation, splicing context, domains, and functional motifs, to understand how sequence changes alter cellular protein products.

I have also contributed to target prioritization in a multi-omics ML project focused on ALS target discovery across 13 transcriptomic datasets (1,227 samples; accepted, npj Syst. Biol. Appl.; wet-lab validation ongoing). The knowledge gained from this project, together with subsequent group discussions, shaped my current work on developing a computational framework to investigate regulatory changes at interface of RNA-level signatures and downstream protein changes in cancer, as both conditions are distant from physiologic homeostasis. Here the aim is to decipher how pleiotropy and multifunctional roles of DNA damage repair proteins influence immune activity and tumor response.

IISER Mohali – Mohali, India | **Project Assistant (Post PhD) |** [07/08/2023 - 15/09/2025]

I continued ENACT and ENACTdb after thesis submission, completed pending analyses, revised the Genome Research (2025) and Bioinformatics Advances (2024) manuscripts based on peer-review feedback, and communicated the framework as a gene-centric resource. The position was funded till January 2025.

Dr. Lal PathLabs – Ludhiana, India| **Trainee - Territorial Manager |** [01/07/2016 - 27/07/2017]

Worked with clinicians in oncology, nephrology, neurology, rheumatology, and gastroenterology to communicate diagnostic tests in molecular biology and biochemistry. The role focused on communicating diagnostic advancements and informing therapeutic regimens for better patient management.

EDUCATION & TRAINING

Ph.D. in Computational Biology |*IISER Mohali* [01/08/2017 - 19/06/2024]

Level in EQF: 8

Thesis: "*Computational insights into alternative splicing driven proteome diversification: An evolutionary perspective*"

The PhD training centered on transcript architecture and its protein-level consequences. I developed ENACT and ENACTdb (Python, Django 4, ReactJS) to classify how alternative transcription, splicing, and translation change exon composition across isoforms in five evolutionarily distant species (*C. elegans*, *D. melanogaster*, *D. rerio*, *M. musculus*, and *H. sapiens*). I then connected those changes to coding sequence and protein features, so that transcript differences could be interpreted additionally to expression level.

During this period I gained experience in biological data modelling, annotation workflows and web-resource development, performed atomistic and steered molecular dynamics (MD) simulations and residue contact-network analyses, administered lab HPC clusters (Slurm, PBS) and NAS storage (Synology, Netgear), and collaborated with three experimental research groups.

I now extend this framework to variant interpretation and proteogenomics, evaluating which DNA- or RNA-level signals alter a functional protein product and extent of functionality (multiple interactions in multi domain proteins).

B. Tech Biotechnology | ***LPU, Phagwara, India*** [01/08/2012 - 31/05/2016] | **CGPA**: 8.31

Level in EQF: 6

PUBLICATIONS

[2026] **Multi-omic machine learning overcomes transcriptomic blind spots to rescue therapeutic targets in ALS.**
Interpreted a 13-dataset cross-tissue ALS meta-analysis (1,227 samples) with an explainable SHAP-decomposed ensemble classifier to prioritize 11 consensus therapeutic candidates. Further validation ongoing at University Hospital Mannheim.
Authors: Gupta M, Verma P, Sharma M. | Accepted: npj Systems Biology and Applications.

[2026] **PEXMap: A proteogenomic method for exon and isoform-level mapping of mass-spectrometry-derived peptides.**
PEXMap enables quantification of genes, transcripts, exons, and exon-exon junctions, from bottom-up mass spectrometry peptides using an 8-mer exact-matching k-mer approach.
Authors: Awasthi D, **Verma P**, Pandit SB. | **Publisher:** *bioRxiv*

[2025] **Exon nomenclature and classification of transcripts (ENACT) provides a systematic framework to annotate exon attributes.**
ENACT enables a structured exon annotation approach across isoforms and connects exon-level variation to protein sequence consequences.
Authors: **Verma P**, Thakur D, Awasthi D, Pandit SB | **Journal Name:** *Genome Research*: 35(6):1440–1455 | **Publisher:** Cold Spring Harbor Laboratory

[2025] **Genome assembly and annotation of the parasitoid jewel wasp *Nasonia oneida*.**
In this work we report the first de novo genome assembly and annotation of Nasonia oneida, a parasitoid jewel wasp.
Authors: Rana A, **Verma P**, Pandit SB, Werren J, Wang X, Raychoudhury R. | **Journal Name:** *Scientific Data.* 1;12(1):1096 | **Publisher:** Springer Nature

[2024] **Exon nomenclature and classification of transcripts database (ENACTdb): a resource for analyzing alternative splicing mediated proteome diversity**
ENACTdb is an accessible database for ENACT framework and allows exploring how changes in exon composition dictate protein features.
Authors: **Verma P**, Thakur D, Pandit SB | **Journal Name:** *Bioinformatics Advances*: 4(1):vbae157 | **Publisher:** Oxford University Press

Protein dynamics publications (2021 - 2025)

- [2025] Deciphering binding site conformational variability of substrate promiscuous and specialist enzymes. Thakur D, Verma P, Pandit SB. *J. Chem. Inf. Model.*, 65(16):8740–8753.
- [2021] Tyrosine in the hinge region of the pore-forming motif regulates oligomeric beta-barrel pore formation by Vibrio cholerae cytolysin. Mondal AK, Verma P, Sengupta N, Dutta S, Pandit SB, Chattopadhyay K. *Molecular Microbiology*. 115(4).
- [2021] Optimal Protein Sequence Design Mitigates Mechanical Failure in Silk β-Sheet Nanocrystals. Verma P, Panda B, Singh KP, Pandit SB. *ACS Biomaterials Science and Engineering*. 7(7).

VOLUNTEERING, ACADEMIC SERVICE, CONFERENCES & SEMINARS

• Mentor for interns and early-stage PhDs: Mentored junior interns and newly joined PhD researchers in computational and scientific skills (Python, R, Bash scripting, logical problem-solving through PDB parsing, and Linux environment management).
• Volunteer & Jury Member, Young Scientist Conclave, IISF 2025: Served as a jury member evaluating posters presented by young scientists, ranging from school students to early-stage PhD researchers.
• POC for IndiaBioscience Exec. Director to Plaksha University: Assisted in coordinating interactions with faculty members.
• Serving currently as department representative for Plaksha's Ambassador programme, which involved orienting young minds towards the research activities, in a simplified manner, through activities.

Conferences:
• Theoretical Chemistry and Biology Symposium. NIPER Mohali, India. 15/10/2022. Best poster award.
• 42nd meeting of the Indian Biophysical Society. IISER Pune, India. March 2018. Poster presentation.
• Biophysics at the Dawn of Exascale Computers thematic meeting. DESY Hamburg, Germany. May 16-20 2022. Poster presentation.

Major talks and seminars
• Fortnightly Data Club series: "BRCAEthniVar: A Multi-Ethnic Framework for Equitable Germline Variant Interpretation". 28/01/2026, Plaksha University, Mohali.
• Department of Biological Sciences Seminar: "Understanding the effect of Alternative Splicing (AS) on protein domains and secondary structure". 18/09/2019, IISER Mohali.
• Science Club Talk: "Gut Microbiota and Host-Immune Crosstalk: Mechanisms and Emerging Breakthroughs" (22/09/2018). Science communication seminar co-led with the 2017 Biology PhD cohort, IISER Mohali.

CREATIVE WORKS

[01/01/2022 - Current] **Interactive full-stack platform design to visualise post-transcriptional processing and gene architecture.** I have designed and developed a full-stack web-based platform to visually represent post-transcriptional processing footprints on gene architecture. This combined illustration of alternative transcription, splicing and translation helps comprehend gene architecture's modular stack. This helped illustrate how modular gene organisation, especially exon structure and alternative usage at transcript ends, contributes to transcript and proteoform diversity. https://www.iscbglab.in/enactdb.

Advancing on this, I am designing a new database to bridge knowledge gap between genetic variant pathogenicity and clinical interpretation, with the aim of supporting clearer functional and biomedical understanding of variants.

[01/06/2024 - 31/05/2026] **Laboratory website design on GitHub pages.** I helped design and develop academic lab websites for doctoral and postdoctoral advisors.